Appendix. Class-wise evaluation of lithology-identification performance for the common component of MWD signals (per-class metrics and confusion matrices)

Section 4 of the main text summarizes the performance of six classifiers on the original signal and on the common component in terms of overall accuracy. This appendix reports class-wise results: precision, recall and F1 for each lithology, macro-averaged F1, balanced accuracy, and the confusion matrices. The protocol is a class-stratified 80/20 split, grid search under 5-fold cross-validation on the training set, and the mean over 10 repeats, with the same random seeds, splitting routine and search grids as in the main text. Because the five lithologies are not equally represented, metrics that weight each class equally are reported here alongside overall accuracy.

1.1 Class distribution of the samples

Table 1. Sample distribution of the five lithologies for each feature combination.

Signal	State	Sandstone A	Sandstone B	Sandstone C	Granite	Limestone	Total
Pressure	Original signal	29	5	11	15	11	71
Pressure	Common component	833	26	105	210	110	1284
Torque	Original signal	31	6	11	16	11	75
Torque	Common component	915	28	127	240	110	1420
Pressure + torque	Original signal	29	6	11	15	11	72
Pressure + torque	Common component	812	30	110	210	110	1272

The lithologies differ in the number of available boreholes, and pairwise pairing expands the sample size in proportion to roughly the square of that number, so lithologies with more boreholes expand faster and the class distribution of the common component is less even than that of the original signal.

1.2 Macro-averaged F1 and balanced accuracy

The two tables below are laid out as Table 9 of the main text, so that they can be compared cell by cell.

Table 2. Macro-averaged F1 of each classifier (mean ± standard deviation over 10 repeats, %).

Signal	State	LDA	KNN	SVM-RBF	XGBoost	LightGBM	Random Forest
Pressure	Original signal	56.2 ± 15.8	65.6 ± 16.3	78.0 ± 13.0	76.4 ± 7.7	74.8 ± 13.8	87.7 ± 10.5
Pressure	Common component	73.7 ± 4.3	88.6 ± 3.8	96.3 ± 1.9	96.7 ± 4.4	97.3 ± 2.9	96.3 ± 2.5
Torque	Original signal	45.5 ± 9.4	59.9 ± 13.0	64.4 ± 16.1	67.1 ± 14.6	56.6 ± 13.6	71.3 ± 9.7
Torque	Common	58.2 ± 3.3	81.9 ± 5.6	90.0 ± 3.8	93.8 ± 1.9	94.8 ± 1.1	94.8 ± 1.8

	component						
Pressure + torque	Original signal	56.4 ± 18.8	66.5 ± 15.3	70.1 ± 14.0	87.1 ± 11.6	72.3 ± 14.8	93.2 ± 8.9
Pressure + torque	Common component	79.4 ± 4.8	89.7 ± 4.5	94.0 ± 2.7	98.5 ± 1.3	99.0 ± 1.2	98.6 ± 0.7

Table 3. Balanced accuracy of each classifier (mean ± standard deviation over 10 repeats, %).

Signal	State	LDA	KNN	SVM-RBF	XGBoost	LightGBM	Random Forest
Pressure	Original signal	60.6 ± 13.4	68.6 ± 15.1	81.6 ± 12.4	79.2 ± 7.6	78.2 ± 12.0	90.2 ± 10.2
Pressure	Common component	71.3 ± 3.4	86.7 ± 4.6	95.6 ± 2.9	95.8 ± 5.4	96.2 ± 4.1	94.6 ± 3.5
Torque	Original signal	52.3 ± 10.8	64.7 ± 13.3	68.3 ± 13.4	71.7 ± 12.8	60.3 ± 13.0	76.0 ± 9.6
Torque	Common component	57.6 ± 3.3	80.4 ± 4.9	88.4 ± 4.0	92.5 ± 2.7	94.4 ± 1.8	93.4 ± 1.9
Pressure + torque	Original signal	63.7 ± 16.6	69.9 ± 13.6	73.3 ± 11.7	87.5 ± 12.0	76.1 ± 15.6	93.5 ± 8.1
Pressure + torque	Common component	80.4 ± 5.1	89.1 ± 5.0	94.4 ± 4.0	97.7 ± 2.1	98.6 ± 1.9	97.6 ± 1.2

Both metrics fall below the corresponding overall accuracy, as expected of equally weighted metrics under unequal class sizes: Sandstone B, the smallest class, carries the same weight as Sandstone A, the largest. The conclusion does not change with the metric. Across all 18 comparisons (3 feature combinations × 6 classifiers), the macro-averaged F1 of the common component exceeds that of the original signal without exception, by 5.4 to 38.2 percentage points. Balanced accuracy likewise improves in all 18 comparisons, by 4.1 to 34.1 percentage points. The repeat-to-repeat standard deviation is also markedly smaller for the common component, that of macro-averaged F1 falling from 7.7–18.8 to 0.7–5.6 percentage points, so classification on common-component features is considerably less sensitive to the particular data split.

1.3 Class-wise metrics and confusion matrices

Table 4 gives the class-wise metrics of the best model for each state under the pressure + torque combination, and Figs. 1 to 3 give the confusion matrices for the pressure, torque and pressure + torque combinations in turn, each using the most accurate model for that combination.

Table 4. Class-wise precision, recall and F1 of the best model under the pressure + torque combination (pooled over 10 repeats, %).

State	Lithology	Support	Precision	Recall	F1
Original signal (Random Forest)	Sandstone A	50	94.1	96.0	95.0
	Sandstone B	10	100.0	90.0	94.7
	Sandstone C	20	100.0	100.0	100.0
	Granite	30	96.7	96.7	96.7
	Limestone	20	85.0	85.0	85.0

Common component (LightGBM)	Sandstone A	1620	99.4	99.9	99.7
	Sandstone B	60	100.0	96.7	98.3
	Sandstone C	220	99.1	96.8	97.9
	Granite	420	100.0	99.8	99.9
	Limestone	220	99.5	100.0	99.8

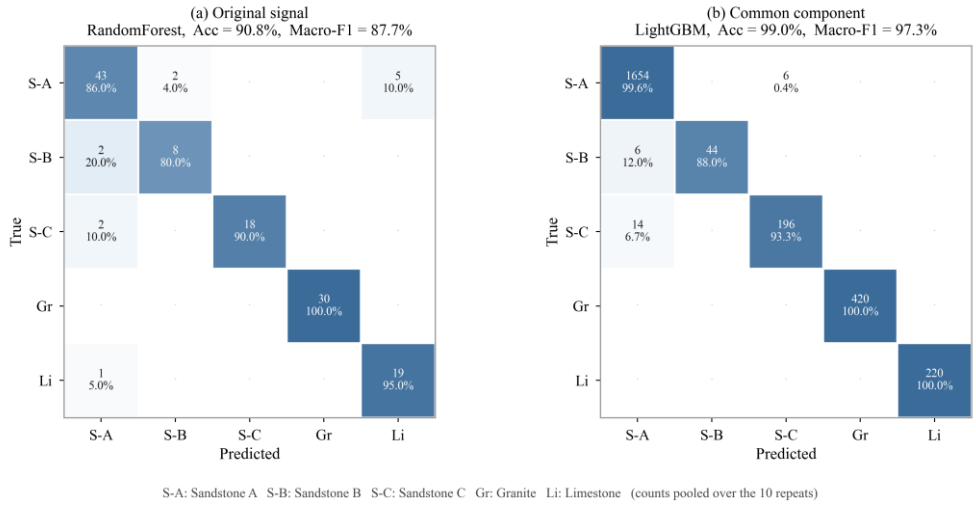


Fig. 1. Confusion matrices for the original signal and the common component under pressure features (pooled over 10 repeats; the best model for that combination is used in each case).

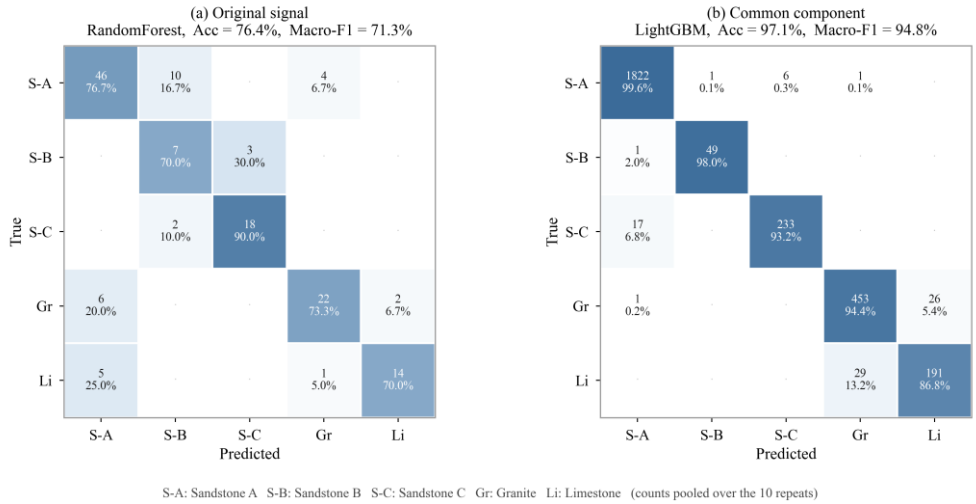


Fig. 2. Confusion matrices for the original signal and the common component under torque features (pooled over 10 repeats; the best model for that combination is used in each case).

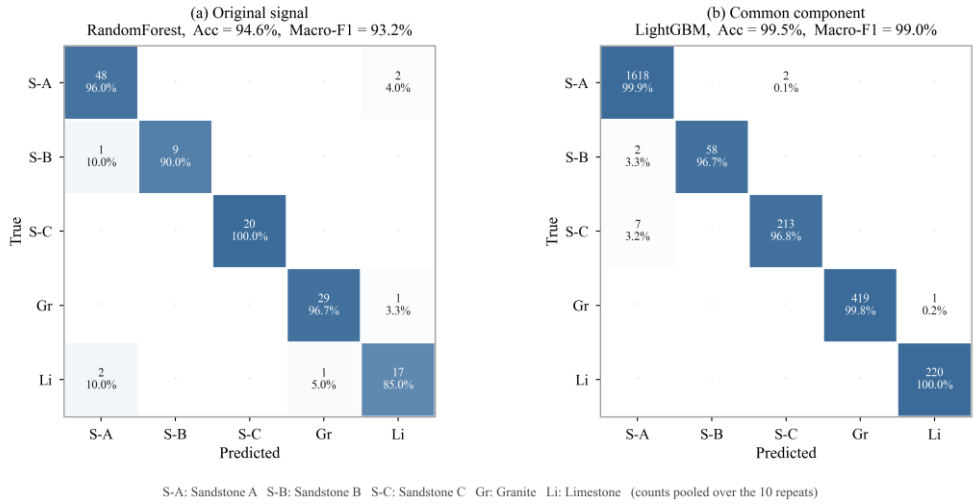


Fig. 3. Confusion matrices for the original signal and the common component under pressure + torque features (pooled over 10 repeats; the best model for that combination is used in each case).

Under the common component the F1 of all five lithologies is at least 97.9%. The lowest belongs to Sandstone C, with precision 99.1% and recall 96.8%, its 7 misclassified samples falling entirely into Sandstone A; Sandstone B, the smallest class, has precision 100.0% and recall 96.7%, its 2 misclassified samples likewise falling into Sandstone A. The remaining errors are few: 2 of the 1620 Sandstone A samples are assigned to Sandstone C, and 1 of the 420 Granite samples to Limestone. On the original signal, by contrast, the recall of Limestone is 85.0% and that of Sandstone B 90.0%, which the common component raises to 100.0% and 96.7%; Sandstone A and Granite change little (96.0% to 99.9% and 96.7% to 99.8%), and Sandstone C, whose 20 original-signal test samples are all recalled, stands at 96.8%.

1.4 Conclusions

Under metrics that weight each class equally, the advantage of the common component over the original signal still holds and is in fact clearer. The macro-averaged F1 improves in all 18 comparisons without exception, by 5.4 to 38.2 percentage points, and balanced accuracy likewise improves in all 18.

The common component also markedly reduces the sensitivity of the results to the data split, the repeat-to-repeat standard deviation of macro-averaged F1 falling from 7.7–18.8 to 0.7–5.6 percentage points, so its discriminative information does not rest on any particular partition of the samples.

The class-wise results show that under the common component the F1 of all five lithologies is at least 97.9%, and that the residual errors are concentrated among the three sandstones: 11 samples exchanged between Sandstone A and Sandstone B or Sandstone C. Outside the sandstones, only a single Granite sample is assigned to Limestone.